

Botanix Reth Security Audit Report (Combined)

Branch	rc
Audit Dates	2026-02-27 and 2026-03-05
Scope	botanix-reth binary + all custom crates
Excluded	node/evm/ (precompiles/engine not active)

This report consolidates two independent audits of the same codebase. Duplicate findings have been merged; where the two reports assigned different severities to the same issue, the higher severity is used.

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ABCI / CometBFT Review

Summary

The ABCI layer is well-structured with strong validation for Non-Deterministic Data (NDD), Bitcoin checkpoint verification, and aggregate public key consistency. However, two Critical issues make the snapshot sync subsystem completely non-functional, and several High-severity panics in consensus-critical paths (`process_proposal`, `finalize_block`) can be triggered by network input of incorrect byte length. A `unreachable!()` macro in both the proposer and validator paths will crash any node that encounters an unrecognized runtime version. One Medium-severity timer bug causes finalization timing metrics to always report zero.

ABCI-001: `offer_snapshot` calls `unimplemented!()` and will always panic

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:1089`
 - **Severity:** Critical
 - **Category:** Panic / Missing implementation
 - **Description:** The entire body of `offer_snapshot` is commented out. The only active code is `unimplemented!()`. CometBFT calls `offer_snapshot` whenever a peer offers a state-sync snapshot; this is a normal protocol operation. Any node configured for state sync, or offered a snapshot by a peer, will immediately panic. Because `spawn_critical` is used for the ABCI server task, this panic propagates and terminates the entire node process.
 - **Suggestion:** Either implement the snapshot offer logic (the commented-out code contains the intended implementation) or return `ResponseOfferSnapshot { result: SnapshotOfferResult::Reject as i32 }` unconditionally until the feature is ready.
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ABCI-002: `apply_snapshot_chunk` silently accepts all chunks without applying any state

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:1548-1552`
 - **Severity:** Critical
 - **Category:** Silent failure / Missing implementation
 - **Description:** All logic inside `apply_snapshot_chunk` is commented out. The function unconditionally returns `ApplySnapshotResult::Accept`. CometBFT will believe chunks are being applied successfully, advance through all chunks, and then call `commit`. The node's state will never have been populated from the snapshot. After a restart, the node will report height 0 despite having "accepted" a full snapshot. Unlike `offer_snapshot`, this does not panic — making the failure completely silent. Even if `offer_snapshot` is later fixed to accept snapshots, chunks would be accepted without being applied, decoded, or validated.
 - **Suggestion:** Return `ApplySnapshotResult::Abort` until the implementation is complete so CometBFT falls back to block sync and does not silently enter a broken state.
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ABCI-003: Unguarded `FixedBytes::<32>::from_slice` on untrusted `request.hash` panics if length `!= 32`

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:1868-1869` (`processproposal`) and `2319-2320` (`finalizeblock`)
- **Severity:** High
- **Category:** Missing bounds check / Panic
- **Description:** Both `process_proposal` and `finalize_block` convert `request.hash` to `FixedBytes::<32>` using `FixedBytes::<32>::from_slice(request.hash.to_vec().as_slice())`. `FixedBytes::from_slice` panics if the input slice is not exactly 32 bytes. `request.hash` is a

`prost::bytes::Bytes` field from the protobuf message and its length is not checked before this call. A malformed or truncated CometBFT message (possible during network faults, upgrades, or adversarial conditions) will crash the validator.

- **Suggestion:** Check the length before conversion and return `VERIFY_REJECT` in `process_proposal` on mismatch:

```
if request.hash.len() != 32 {  
    error!("Invalid block hash length: {}", request.hash.len());  
    return ResponseProcessProposal { status: VERIFY_REJECT };  
}
```

ABCI-004: `Address::from_slice` on untrusted `request.proposer_address` panics if length != 20

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:2508` and `2390-2393`
- **Severity:** High
- **Category:** Missing bounds check / Panic
- **Description:** In `finalize_block`, `Address::from_slice(&request.proposer_address)` panics if the slice is not exactly 20 bytes. CometBFT validator addresses are Ed25519 public key hashes (20 bytes), but the `proposer_address` field is typed as `bytes` in protobuf — there is no guarantee it will be exactly 20 bytes. The same unchecked `FixedBytes::<20>::from_slice` issue exists on the testnet historical-sync path at lines 2390-2393. A CometBFT misconfiguration or protocol error sending an address of different length causes a node crash in `finalize_block`.
- **Suggestion:** Validate the length before conversion and reject with a clear diagnostic message:

```
let proposer_address: Address = request.proposer_address  
    .as_ref()  
    .try_into()  
    .map_err(|_| { error!("proposer_address must be 20 bytes"); return VERIFY_REJECT; })?;
```

ABCI-005: `unreachable!()` in proposer and validator paths crashes nodes on unrecognized `RuntimeVersion`

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:1612` (`prepareproposal`) and `2128` (`processproposal`)
 - **Severity:** High
 - **Category:** Panic / Forward compatibility
 - **Description:** Both `prepare_proposal` and `process_proposal` match on `RuntimeVersion` and use `_ => unreachable!()` for any version not in V1-V3. When the network upgrades to V4+, older nodes will receive blocks with the upgraded runtime version and will panic at this branch during `process_proposal`, crashing the validator. Notably, `finalize_block` handles this correctly with a `warn!` + fallback to the latest known floor fee — creating an intentional but undocumented inconsistency between the three paths.
 - **Suggestion:** Replace `unreachable!()` with `warn!` + `VERIFY_REJECT` in `process_proposal` and `warn!` + fall-through to the latest known floor fee in `prepare_proposal`, consistent with how `finalize_block` already handles this case.
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ABCI-006: Integer underflow `txs_len - 1` panics in debug builds when proposal has zero transactions

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:1895` (and ~8 additional sites throughout `process_proposal`)
 - **Severity:** High
 - **Category:** Panic / Integer underflow
 - **Description:** When `request.txs` is empty (`txs_len == 0`), the code reaches the `None` branch of `txs_bytes.first()` and attempts to log `eth_transactions_count = txs_len - 1` — `0usize - 1` panics in debug builds. In release builds it wraps to `usize::MAX`, producing a nonsensical log value. The same pattern appears at least 8 times within `process_proposal`. The panic occurs inside a `tracing::enabled!(tracing::Level::WARN)` guard, so it triggers whenever WARN-level logging is enabled — which it almost always is in production.
 - **Suggestion:** Use `txs_len.saturating_sub(1)` throughout, or compute `eth_transactions_count` as a guarded value before the log sites.
-

ABCI-007: Unchecked `i64` to `u64` cast of `request.height`

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:1583, 2100, 2506`
- **Severity:** Medium
- **Category:** Integer overflow / Data integrity
- **Description:** CometBFT protobuf defines `height` as `int64` (i64 in Rust). The code casts it with `request.height as u64` without validating that the value is non-negative. While CometBFT should never send a negative height in normal operation, a malformed or

adversarial CometBFT instance could send a negative value, which would wrap to a very large u64 and be passed to `activation_manager.on_prepare_proposal()`, `on_process_proposal()`, and `on_finalize_block()`, potentially causing incorrect activation decisions.

- **Suggestion:** Add a guard at the start of each handler that validates `request.height > 0` and use `try_into()`:

```
let comet_height: u64 = request.height

    .try_into()

    .expect("CometBFT height must be positive");
```

ABCI-008: `finalize_block` execution timer always reports zero seconds

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:2637`
- **Severity:** Medium
- **Category:** Logic error / Incorrect metrics
- **Description:** The execution time is computed as `std::time::Instant::now().elapsed().as_secs_f32()` — a newly created `Instant` immediately calls `.elapsed()`, which always returns approximately zero. The start of the function is never timestamped, so the logged `execution_time` in the `finalize_block` info! line will always be ~0.0 seconds, making the performance metric useless and masking potential performance regressions.
- **Suggestion:** Add `let execution_start_time = std::time::Instant::now();` at the start of `finalize_block` (analogous to `prepare_proposal` line 1561) and replace line 2637 with `let execution_time = execution_start_time.elapsed().as_secs_f32();`.

ABCI-009: `remove_persisted_blocks` called with mismatched `NumHash` (wrong hash for the given height)

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:3006-3011`
- **Severity:** Medium
- **Category:** Logic error / Incorrect state management
- **Description:** After committing a block, the in-memory canonical state is pruned with `NumHash::new(block_height - 1, new_header.hash_slow())`. The height is `block_height - 1` (the previous block) but the hash is `new_header.hash_slow()` (the current/new block). These two values belong to different blocks, forming an internally inconsistent `NumHash`. Depending on how `remove_persisted_blocks` uses this value, the wrong blocks may be retained or pruned from in-memory state.

- **Suggestion:** Pass the previous block's hash: use `new_header.parent_hash` instead of `new_header.hash_slow()`.
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ABCI-010: `expect("db cannot fail")` in activation manager calls

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:1584, 2108, 2519, 2542`
 - **Severity:** Medium
 - **Category:** Error handling
 - **Description:** Multiple calls to the activation manager use `.expect("db cannot fail")`. Disk full, I/O errors, or database corruption could cause these to panic. In `prepare_proposal` (line 1584), this panic kills the proposer. In `process_proposal` (line 2108), it kills a validator during proposal verification; a more graceful response would be to reject the proposal. In `finalize_block` (lines 2519, 2542), panicking may be arguably appropriate since the node cannot proceed with inconsistent state, but should use structured shutdown rather than a bare panic.
 - **Suggestion:** In `process_proposal`, replace `expect` with proper error handling that returns `VERIFY_REJECT`. In `prepare_proposal`, return a graceful error rather than crashing.
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ABCI-011: `blocking_read()` / `blocking_write()` on async `RwLock` in async context

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:663, 2576`
 - **Severity:** Medium
 - **Category:** Concurrency / Deadlock risk
 - **Description:** `blocking_read()` and `blocking_write()` on a `tokio::sync::RwLock` block the current OS thread. When called from an async context (which ABCI handlers are), this can block the Tokio runtime thread, preventing other tasks from making progress, and can cause deadlocks if the lock holder is waiting on a task that needs the blocked thread.
 - **Suggestion:** Use `.read().await` and `.write().await` instead, which cooperatively yield to the executor while waiting.
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ABCI-012: Non-deterministic data in `payload_builder_arguments` (fragile override assumption)

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:558-561`
- **Severity:** Low
- **Category:** Consensus safety
- **Description:** `payload_builder_arguments` uses `unix_timestamp()` (system time) and `FixedBytes::<32>::random()` for `prev_randao`. Code comments state these are overridden by CometBFT values during `build_and_execute()`. The reliance on these values being overridden is implicit and fragile — any refactoring that uses `PayloadAttributes` directly without the override step could introduce non-determinism causing consensus splits.

- **Suggestion:** Add an explicit comment or assertion at the `build_and_execute` call site. Consider using sentinel values (e.g., `B256::ZERO`) to make it obvious if the override is ever missed.
-

ABCI-013: Potential `u64` underflow in `remove_persisted_blocks` at block height 0

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:3009`
 - **Severity:** Low
 - **Category:** Integer overflow
 - **Description:** `block_height - 1` is computed where `block_height` is `u64`. If `block_height` is 0 (genesis), this would underflow to `u64::MAX`. In practice genesis is unlikely to flow through this commit path, but it is a defensive programming concern.
 - **Suggestion:** Use `block_height.saturating_sub(1)` or add an assertion that `block_height > 0`.
-

ABCI-014: NDD Version 1 silently upgrades to Version 2 fields during deserialization

- **File:** `bin/botanix-reth/src/consensus/comet_bft/non_deterministic_data.rs:252-258`
 - **Severity:** Low
 - **Category:** Data integrity / Consensus safety
 - **Description:** When deserializing NDD with `VERSION_1`, the code optionally parses Version 2 fields (`runtimeversion`, `networkupgrade_payload`) from remaining bytes. If successful, these fields are applied to a Version 1 NDD struct but the `version` field remains `VERSION_1`. This creates an ambiguity where a Version 1 NDD can silently carry Version 2 data, potentially causing validators to process different runtime versions depending on how they interpret the NDD.
 - **Suggestion:** Either reject trailing bytes for Version 1 (strict parsing), or change the version field to `VERSION_2` when the optional fields are successfully parsed. Document the backward-compatibility rationale explicitly.
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ABCI-015: Block cache limited to 5 entries may cause redundant re-execution under adversarial conditions

- **File:** `bin/botanix-reth/src/consensus/comet_bft/abci.rs:503`
- **Severity:** Low
- **Category:** State consistency / Performance
- **Description:** The block cache uses `ByLength::new(5)`. If more than 5 proposals are processed before any is finalized (e.g., due to a Byzantine proposer flooding proposals), earlier entries are evicted. When `finalize_block` is called for an evicted block it re-

executes from scratch — the fallback is correct, but this can be exploited for performance degradation.

- **Suggestion:** Consider making the cache size configurable or increasing it. The current fallback is correct, so this is a performance concern rather than a safety issue.

Block Execution Review

Summary

The block execution layer has generally sound architecture. However, there is one Critical issue: coordinator selection is hardcoded to always use index 0, making it a single point of failure. Several High-severity panics exist in execution-critical paths. A timestamp integer narrowing cast can produce non-deterministic block timestamps. A pegin amount check uses the wrong comparison operator, rejecting valid exact-match pegins. The signing state machine has TOCTOU races and incomplete cleanup paths.

EXEC-001: Hardcoded coordinator selection — `is_inturn()` and `current_inturn_index()` always return index 0

- **File:** `crates/botanix-consensus-common/src/utils.rs:58-83`
- **Severity:** Critical
- **Category:** FROST signing / Consensus integrity
- **Description:** `is_inturn` always returns `signer_index == 0` and `current_inturn_index` always returns `0`, ignoring all parameters (`authorities_len`, `time_range`, `random_source`). A `TODO: change to round robin` comment confirms this is a placeholder that shipped to production. This means: (1) only authority at index 0 ever acts as the signing coordinator; (2) if authority 0 goes offline, no signing sessions can be initiated — all pegouts stall; (3) `RandomSourceProvider::random_source()` also returns all zeros, making the "random" source deterministic and useless. This is a single point of failure for the entire Bitcoin peg-out mechanism.
- **Suggestion:** Implement proper round-robin or epoch-based coordinator rotation, e.g., `block_number % signers_count`. Ensure the random source is backed by a proper CSPRNG, not a constant zero value.

EXEC-002: Panics in block execution critical path via `expect` / `panic!`

- **File:** `bin/botanix-reth/src/consensus/excecution_utils.rs:273-278, 285, 339, 409, 491`
- **Severity:** High
- **Category:** Panic in production execution path

- **Description:** Multiple `expect()` and `unwrap_or_else(|| panic!(...))` calls exist in the block building and execution path — a hot path that runs on every block. If any of these database lookups fail (database corruption, concurrent pruning, first-time genesis), the node panics and crashes rather than returning an error. The project's clippy configuration explicitly disallows `panic!` outside of tests. Key sites:
 - Line 273-278: `unwrap_or_else(|| panic!(...))` if best block hash is not found.
 - Line 285: `.expect("header to exist")` when fetching parent header for base fee.
 - Line 339: `.expect("header to be found")` when fetching parent for Cancun logic.
 - Line 409: `.expect("parent hash exists")` when getting history by block hash for state root.
 - Line 491: `.expect("parent hash exists")` in the `execute` function.
 - **Suggestion:** Replace all `expect()/panic!()` calls with proper error propagation using `?` and `BlockExecutionError::Internal(...)`. The functions already return `Result`, so this is straightforward.
-

EXEC-003: Timestamp truncation silently produces incorrect block timestamps

- **File:** `bin/botanix-reth/src/consensus/excecution_utils.rs:280`
 - **Severity:** High
 - **Category:** Integer narrowing / Non-deterministic execution
 - **Description:** The CometBFT `Timestamp.seconds` field is an `i64`. Casting it directly to `u64` with `as u64` is silent on negative values. A negative `seconds` value (misconfigured CometBFT clock, or a compromised proposer) wraps to a very large `u64`, producing a wildly incorrect block timestamp. This timestamp is then used to select EIP-1559 base fee parameters and determine active hardforks, meaning two nodes that receive slightly different timestamp values could produce different state roots, splitting consensus. The same `as u64` pattern on `request.height` at `abci.rs:1583, 2100, 2506` presents the same risk.
 - **Suggestion:** Use `u64::try_from(timestamp.seconds).map_err(...)` and validate that the timestamp is non-negative and monotonically increasing relative to the parent block.
-

EXEC-004: Pegin amount comparison is inverted — rejects valid exact-match pegins

- **File:** `crates/botanix-evm/src/execute.rs:669-679`
- **Severity:** High
- **Category:** Logic error in peg-in financial validation
- **Description:** The check `if pegin_data.amount >= aggregate_value` rejects the pegin if the requested mint amount is *at least as large* as what was deposited. This means a user who deposits exactly 1 BTC and requests exactly 1 BTC minted is rejected (`amount ==`

aggregate_value is true), while a user requesting 0.999 BTC for a 1 BTC deposit is accepted. The intent appears to be `amount > aggregate_value` to reject over-minting.

- **Suggestion:** Change the check to `amount > aggregate_value` if the intent is to reject over-minting only. Document the fee policy explicitly rather than encoding it as an implicit strict-less-than.

EXEC-005: Panic in V1 pegin database lookup crashes node instead of reverting transaction

- **File:** `crates/botanix-evm/src/execute.rs:612`
- **Severity:** High
- **Category:** Panic in production execution path
- **Description:** Inside `botanix_mint_contract_checks`, a database error when fetching the reference block hash for V1 pegins causes an unconditional `panic!` rather than a graceful error that would cause the transaction to revert. Any database error during block execution will terminate the node rather than fail the transaction. This code path is reached for every V1 pegin transaction included in a block being executed.
- **Suggestion:** Return `MintContractError::InvalidPeginData { ... }` so the transaction is cleanly reverted, consistent with all other error branches in the same function.

EXEC-006: `expect()` calls in `post_block_balance_increments` fee distribution

- **File:** `crates/botanix-evm/src/state.rs:42-79`
- **Severity:** High
- **Category:** Panic in production execution path
- **Description:** Multiple `expect()` calls in the block fee distribution path (`.expect("FeeReceiver to exist")`, `.expect("Valid FeeReceiver")`, `.expect("overflow incrementing balance for LST FeeReceiver")`) will panic on every block if the chain spec has a malformed address string. The outer `is_some()` check ensures the Option is Some, but `Address::from_str(...)` can still fail if the string is not a valid Ethereum address.
- **Suggestion:** Validate fee recipient addresses at startup and cache parsed `Address` values in the chain spec, or propagate parsing errors with `?`.

EXEC-007: Uninitialized variable `total_expected_chunks` read before first assignment

- **File:** `bin/botanix-reth/src/consensus/frost_task.rs:201, 256`
- **Severity:** Medium
- **Category:** Logic error / Uninitialized variable
- **Description:** `total_expected_chunks` is declared with `let mut total_expected_chunks;` (no initializer) and is only assigned inside the `Ok(...)` arm of the streaming loop. If the first

item from the stream is an `Err`, the code falls into the `continue` branch, then reaches the comparison `if (received_healthy_chunks == total_expected_chunks)` — reading an uninitialized variable. The completion check compares against stale data from a prior iteration when any error chunk is received first.

- **Suggestion:** Initialize to `0u64`. Add a guard so the completion check is only evaluated when at least one successful chunk has been processed.

EXEC-008: `is_coordinator` evaluated at different timestamps — coordinator can change mid-session

- **File:** `bin/botanix-reth/src/consensus/signing.rs:536-538, 472-479`
- **Severity:** Medium
- **Category:** TOCTOU / Race condition in signing coordinator selection
- **Description:** `is_coordinator()` internally calls `unix_timestamp()`. In `initiate_signing_session`, the coordinator determination and the downstream check in `signer_process_round1` and `coordinator_process_round1` are evaluated at different wall-clock moments. If an epoch boundary falls between these calls, one side will believe it is not the coordinator and drop the message silently, causing liveness failure.
- **Suggestion:** Capture the coordinator determination once per signing session at session initiation and pass it through. Do not re-evaluate `is_coordinator()` mid-session.

EXEC-009: `.expect` on runtime_version DB read panics on startup

- **File:** `bin/botanix-reth/src/consensus/builder.rs:146-148`
- **Severity:** Medium
- **Category:** Panic during node startup
- **Description:** During node startup in `try_new`, `.expect("local db must be available")` and related expects (lines 147, 219, 303, 322) will panic if the database returns an error or if construction fails. A corrupt or inaccessible database at startup should produce a descriptive error rather than an unrecoverable panic.
- **Suggestion:** Propagate errors using `?` and provide context via `.context("...")` or `.wrap_err("...")`, returning `Err(AuthorityConsensusBuilderError::InvalidStorage)`.

EXEC-010: `.expect` on `effective_tip_per_gas` in fee computation

- **File:** `crates/botanix-evm/src/execute.rs:386-388`
- **Severity:** Medium
- **Category:** Panic in production execution path
- **Description:** `transaction_fee` calls `.expect("base fee exists")`, which will panic if `effective_tip_per_gas` returns `None` — which it can for legacy transactions when `base_fee` is `None`.

- **Suggestion:** Replace `.expect("base fee exists")` with a proper error return using `ok_or(BlockExecutionError::...)`.
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EXEC-011: `is_block_age_acceptable` uses wall-clock time — non-deterministic across nodes

- **File:** `bin/botanix-reth/src/consensus/utils.rs:400-408`
 - **Severity:** Medium
 - **Category:** Non-deterministic execution
 - **Description:** `is_block_age_acceptable` calls `SystemTime::now()` to compare against block timestamps. If this function is ever used on the block execution or validation path, different nodes with different wall-clock times will produce different results, splitting consensus. Currently appears to be on the wallet-sync path only, but this is not clearly documented.
 - **Suggestion:** Confirm this function is only called on the wallet-sync (off-chain coordination) path. Add a comment making this explicit. For any on-chain path, use block timestamps exclusively.
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EXEC-012: Wallet state streaming to peers without federation membership check

- **File:** `bin/botanix-reth/src/consensus/frost_task.rs:698-743`
 - **Severity:** Medium
 - **Category:** Missing validation / Potential DoS
 - **Description:** When a `WalletState` message arrives with empty `finalized_pegout_ids`, the frost task responds by streaming the entire finalized pegout ID set to the requesting peer without verifying that the peer is an authenticated federation member. Any connected peer can trigger a potentially expensive streaming RPC to the btc-server and receive the full finalized pegout list.
 - **Suggestion:** Before responding, verify the requesting `peer_id` is a known federation member (present in `frost_config.authorities` or a connected peer whose frost identifier is known).
-

EXEC-013: `blob_gas_used` computed then unconditionally overwritten to zero

- **File:** `bin/botanix-reth/src/consensus/excecution_utils.rs:292-303, 343`
- **Severity:** Medium
- **Category:** Block validation / Execution correctness
- **Description:** At lines 292-303, `blob_gas_used` is computed by iterating over all transactions and summing their blob gas. However, at line 343 (inside the Cancun activation block) it is unconditionally overwritten to `Some(0)`. Any blob transactions in the

block would have their gas usage ignored in the header, creating an inconsistency between the actual blob gas consumed and the value reported in the header.

- **Suggestion:** Remove the overwrite at line 343, or clarify with a comment that Botanix intentionally does not support blob transactions and add validation to reject blob transactions earlier in the pipeline.

EXEC-014: State root computed but result discarded — dual computation risks divergence

- **File:** `bin/botanix-reth/src/consensus/excecution_utils.rs:181-187`
- **Severity:** Medium
- **Category:** Execution correctness
- **Description:** In `build_and_execute`, `StateRoot::overlay_root_with_updates` is called and the result `_state_root` is intentionally discarded. The state root was already set in `complete_header` via `state_root()`. If the two state root computations diverge (e.g., due to different input states), the trie updates would be inconsistent with the header's state root. This dual computation also wastes CPU time on an expensive operation.
- **Suggestion:** Either verify that `_state_root` matches `completed_header.state_root` (with an assertion), or refactor to compute the state root and trie updates in a single place to avoid divergence.

EXEC-015: Missing PSBT validation that round 2 data originates from the expected session

- **File:** `bin/botanix-reth/src/consensus/signing.rs:606-703, 795-875`
- **Severity:** Medium
- **Category:** FROST signing
- **Description:** In `signer_process_round1` and `signer_process_round2`, the signer validates that the message comes from the coordinator (by checking `frost_identifier`), but the raw PSBT bytes are forwarded directly to the btc-server without verifying they match the original PSBT from session initiation. A malicious coordinator could send a different PSBT in round 2 than in round 1, potentially tricking signers into signing a different transaction. The `original_psbt` field exists in `SigningSession` but is only stored for the coordinator and is never read or validated anywhere in the signing flow.
- **Suggestion:** Store the original PSBT hash when the signer first receives it in round 1. In round 2, verify the PSBT corresponds to the same transaction by comparing the unsigned transaction hash.

EXEC-016: Signing sessions never cleaned up on successful finalization

- **File:** `bin/botanix-reth/src/consensus/signing.rs:795-875, 878-941`
- **Severity:** Medium

- **Category:** FROST signing / Denial of service
 - **Description:** In `coordinator_process_round2`, when signing is finalized the session state is updated to `Finalized` but the session is never removed from `signing_states`. `remove_signing_session` is only called in error paths. For non-coordinator signers, sessions that reach Round2 are also never cleaned up. Over time, `signing_states: HashMap<[u8; 32], SigningSession>` grows unboundedly, consuming memory proportional to the number of signing sessions that have ever occurred.
 - **Suggestion:** Add `self.remove_signing_session(session_id).await;` after successful finalization in `coordinator_process_round2`. Add cleanup for signers after they complete round 2. Consider a TTL-based eviction mechanism for stale sessions.
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EXEC-017: TOCTOU race in signing state machine session checks

- **File:** `bin/botanix-reth/src/consensus/signing.rs:176-218`
 - **Severity:** Medium
 - **Category:** Concurrency
 - **Description:** `update_signing_state`, `insert_new_signing_session`, and related methods acquire the `RwLock` for a read, check a condition, release the lock, then acquire a write lock to perform the mutation. Between the `read()` and `write()` calls, another task could insert the same session ID, leading to duplicate session creation. While the current architecture appears single-task, `SigningStateMachine` is built with `Arc<RwLock<...>>` implying shared access design intent.
 - **Suggestion:** Use a single `write()` lock for check-then-act patterns, or use the `entry()` API on the `HashMap` within a single write lock acquisition.
-

EXEC-018: Panics in builder/frost_task during DKG and startup

- **File:** `bin/botanix-reth/src/consensus/frost_task.rs:493-500, 513, 1025, 1101`
 - **Severity:** Medium
 - **Category:** Error handling
 - **Description:** Several `panic!` and `expect` calls in the frost task can crash the node: `panic!("Error getting receiver handle...")` and `panic!("Failed to send GetPeerMessagesStream...")` in `start_task`; `.expect("invalid aggregated public key")` when hex-decoding the public key from btc-server; `.expect("valid frost id")` in DKG gossip. A malformed response from btc-server (e.g., invalid hex in public key) would crash the entire node. The DKG runner runs in a tokio task, so a panic kills that task silently.
 - **Suggestion:** Replace panics with proper error handling. For `start_task`, return an error or log and retry. For DKG, use `match / ?` and log errors instead of panicking.
-

EXEC-019: Snapshot manager holds `std::sync::RwLock` across async boundaries

- **File:** `bin/botanix-reth/src/consensus/snapshot_manager.rs:366-371, 395-396, 494-539`
 - **Severity:** Medium
 - **Category:** Concurrency / Deadlock risk
 - **Description:** `SnapshotManagerStateLock` uses `std::sync::RwLock` (not `tokio::sync::RwLock`). In `process_block`, the write guard is held while performing I/O operations. Holding a `std::sync::RwLock` guard across `.await` points would block the tokio runtime thread. Multiple `expect("snapshot state sync locked")` calls on lock acquisition will panic on poisoning.
 - **Suggestion:** Migrate to `tokio::sync::RwLock` for consistency with the rest of the codebase, or carefully document that no `.await` points exist while the guard is held. Replace `.expect("snapshot state sync locked")` with proper error handling.
-

EXEC-020: Builder uses `unwrap()` on CometBFT RPC connection in health check

- **File:** `bin/botanix-reth/src/consensus/builder.rs:389`
 - **Severity:** Low
 - **Category:** Error handling
 - **Description:** In the health monitoring task setup, `cometbft_rpc_factory.build_and_connect().unwrap()` is called. If the CometBFT RPC is unavailable at startup, this will panic in `spawn_critical`, potentially taking down the entire node.
 - **Suggestion:** Handle the error gracefully and retry the connection inside the monitoring loop, or log and skip the CometBFT health check if the initial connection fails.
-

EXEC-021: Infinite retry loop in `get_latest_comet_block_height`

- **File:** `bin/botanix-reth/src/consensus/snapshot_manager.rs:613-624`
 - **Severity:** Low
 - **Category:** Denial of service
 - **Description:** `get_latest_comet_block_height` retries forever with a 2-second delay if the CometBFT RPC is unavailable. If the RPC server never comes online (misconfiguration, network partition), the snapshot manager task will loop indefinitely.
 - **Suggestion:** Add a maximum retry count or exponential backoff with a cap, and return an error after exhausting retries.
-

EXEC-022: Wallet state sync spawns unbounded long-lived task with no cancellation

- **File:** `bin/botanix-reth/src/consensus/wallet_state_sync.rs:233-415`
 - **Severity:** Low
 - **Category:** Concurrency / Resource management
 - **Description:** `sync_wallet_state` spawns a background task that loops indefinitely with no cancellation mechanism (`CancellationToken` or shutdown signal). If called multiple times due to reconnection, multiple background tasks accumulate, each processing the same peer messages.
 - **Suggestion:** Use a `tokio_util::sync::CancellationToken` or `tokio::sync::watch` channel to signal shutdown. Ensure only one background listener is active at a time.
-

EXEC-023: `generate_signing_session_id` uses UUID hex bytes, not raw random bytes

- **File:** `bin/botanix-reth/src/consensus/utils.rs:519-527`
 - **Severity:** Low
 - **Category:** Cryptographic issues / Code quality
 - **Description:** The function generates a UUID v4, converts it to a hex string, then takes the ASCII bytes of that hex string as the session ID. The 32-byte session ID only contains values in the ASCII hex range, reducing the effective entropy and creating session IDs that are visually confusing (hex-of-hex). The `parse_signing_session_id` function also checks `if session_id.len() != 32` on a `&FixedBytes<32>`, which always has a length of 32 — the check can never fail and provides false confidence.
 - **Suggestion:** Use `Uuid::new_v4().into_bytes()` directly, or `rand::thread_rng().gen:::<[u8; 32]>()` for full entropy. Remove the redundant length check from `parse_signing_session_id`.
-

EXEC-024: `find_epoch_start` uses subtraction loop instead of modular arithmetic

- **File:** `bin/botanix-reth/src/consensus/utils.rs:280-289`
 - **Severity:** Info
 - **Category:** Performance
 - **Description:** `find_epoch_start` decrements `start_block_number` in a loop until it is divisible by `epoch_length`. This is $O(\text{epoch_length})$ and could loop indefinitely if `epoch_length` is 0.
 - **Suggestion:** Replace with `current_block_number - (current_block_number % epoch_length)` for $O(1)$ performance. Add a guard for `epoch_length == 0`.
-

Node Services Review

Summary

The node services layer is generally well-structured with good use of typed errors, fallback mechanisms, and task supervision. However, several security-relevant issues were identified: Bitcoin block headers accepted from RPC are not validated for proof-of-work; the BTC server JWT secret is silently optional, allowing unauthenticated connections to the signing service; all RPC namespaces (including `admin` and `debug`) are unconditionally exposed on all transports; and the light client builder panics unconditionally on connection failure at startup.

SVC-001: Bitcoin Block Headers Accepted Without Proof-of-Work Validation

- **File:** `crates/botanix-bitcoin-checkpoint/src/syncer.rs:212-231`
 - **Severity:** High
 - **Category:** Missing validation / Checkpoint integrity
 - **Description:** In `sync_new_blocks`, the syncer fetches block headers from bitcoind via RPC and stores them as checkpoints. The only structural check performed is the parent-hash linkage enforced in `BitcoinCheckpointsChain::push`. There is no validation that each header's proof-of-work is valid (i.e., that the block hash meets the `nBits` difficulty target). A compromised or malicious bitcoind — or a man-in-the-middle on an unencrypted RPC connection — could serve fabricated headers with arbitrary content. Because the chain only checks hash linkage, an attacker controlling the RPC layer could feed a fully fabricated linear chain, making the checkpoint used for peg-in/peg-out decisions attacker-controlled.
 - **Suggestion:** After fetching each header, validate that `header.block_hash()` satisfies the difficulty target using the bitcoin crate's `Header::validate_pow`. Additionally, ensure bitcoind RPC connections are made over TLS or an authenticated local socket.
-

SVC-002: BTC Server JWT Authentication Is Silently Optional in Federation Mode

- **File:** `bin/botanix-reth/src/services/btc_server.rs:17-18`
- **Severity:** High
- **Category:** Missing authentication
- **Description:** The JWT secret for the BTC signing server is loaded via `bitcoind_cfg.btc_signing_server_jwt_secret().ok().flatten()`. The `.ok()` call silently converts any `JwtError` (file not found, invalid format) into `None`, and `.flatten()` passes `None` as the JWT secret — connecting without authentication. No error is surfaced when the secret is missing or misconfigured. Additionally, `bitcoind_cfg.btc_server.clone().expect("btc_server exists")` will panic if `btc_server`

config is `None` when federation mode is enabled. The BTC signing server performs FROST threshold signing over Bitcoin transactions, making unauthenticated connections a critical security gap.

- **Suggestion:** In federation mode the JWT secret must be mandatory. Replace `.ok().flatten()` with proper error propagation:

```
let jwt_secret = bitcoind_cfg.btc_signing_server_jwt_secret()
    .map_err(|e| eyre::eyre!("Failed to load BTC server JWT secret: {}", e))?
    .ok_or_else(|| eyre::eyre!("BTC server JWT secret path is required in federation mode"))?;
```

Replace the `expect("btc_server exists")` with

```
bitcoind_cfg.btc_server.clone().ok_or_else(|| eyre::eyre!("btc_server URL must be
configured when federation_mode is enabled"))?.
```

SVC-003: All RPC Namespaces Exposed on All Transports Without Restriction

- **File:** `bin/botanix-reth/src/services/rpc/rpc.rs:71-74`
- **Severity:** High
- **Category:** Missing authorization / Over-exposure
- **Description:** The RPC server is configured with `RethRpcModule::all_variants()` on HTTP, WebSocket, and IPC transports. This includes the `admin` namespace (peer management, node info, data dir paths) and `debug` namespace (state dumps, tracing, raw RLP, and `setHead` to rewind the chain). There is no role-based filtering, no IP allowlist, and no separation for privileged namespaces. A node operator binding HTTP/WS RPC to a public interface exposes admin and debug APIs to any caller. `debug_setHead` can rewind the chain to an arbitrary block with a single unauthenticated call.
- **Suggestion:** Restrict dangerous namespaces to the IPC transport only:

```
let public_modules = [RethRpcModule::Eth, RethRpcModule::Net, /* ... */];

let module_config = TransportRpcModuleConfig::default()

    .with_http(public_modules)

    .with_ws(public_modules)

    .with_ipc(RethRpcModule::all_variants());
```

SVC-004: Light Client Builder Panics on RPC Failure at Startup

- **File:** `crates/botanix-comet-bft-rpc/src/light_client.rs:45-82`

- **Severity:** Medium
 - **Category:** Panic in service-critical path
 - **Description:** `LightCBFTClientBuilder::build_and_verify`, called during node startup, uses `.expect()` and unconditional `panic!` on connection failure, `status()` failure, block fetch failure, and light client build failure. Any transient network failure connecting to CometBFT RPC during startup crashes the process instead of returning a graceful error.
 - **Suggestion:** Change the return type to `eyre::Result<Instance>` and propagate errors using `?` rather than `.expect()` or `panic!`.
-

SVC-005: Panics in FROST task on channel errors

- **File:** `bin/botanix-reth/src/consensus/frost_task.rs:493, 499`
 - **Severity:** Medium (see also EXEC-018)
 - **Category:** Panics in production
 - **Description:** Channel errors in the FROST signing task (`panic!("Error getting receiver handle")`, `panic!("Failed to send GetPeerMessagesStream frost command")`) cause immediate node crash. During network partitions or when peers disconnect, these channel errors can occur legitimately.
 - **Suggestion:** Handle channel errors gracefully by logging and restarting the signing session rather than crashing.
-

SVC-006: `is_synced` Silently Converts RPC Errors to `false`

- **File:** `crates/botanix-btc-wallet/src/bitcoind.rs:87-93`
 - **Severity:** Medium
 - **Category:** Silent failure / Error handling
 - **Description:** The `is_synced` implementation silently maps any RPC error to `Ok(false)`. A node that has lost its RPC connection to bitcoind will appear perpetually "not synced" rather than returning an error. The `wait_until_synced` loop will spin forever polling non-functional bitcoind at 5-second intervals. More critically, `Ok(false)` is not an `Err` — the `FallbackBitcoindClient` is bypassed because it never gets a chance to serve responses.
 - **Suggestion:** Return `Err(BitcoindError::BlockchainInfoFailed(e))` from the `Err` arm. The `wait_until_synced` loop should distinguish between "not yet synced" (`Ok(false)`) and "RPC failure" (`Err`), falling back to the secondary client on failure.
-

SVC-007: `trigger_checkpoints_sync` Panics When Channel Is Closed

- **File:** `crates/botanix-bitcoin-checkpoint/src/syncer.rs:339-343`
- **Severity:** Medium
- **Category:** Panic in background task

- **Description:** `trigger_checkpoints_sync` panics unconditionally if the internal mpsc channel is closed. This function is called from a bare `tokio::spawn` task (not `task_executor.spawn_critical`). A panic in a bare `tokio::spawn` task aborts the task silently — the checkpoints synchronizer will stop updating permanently without any operator-visible critical error.
 - **Suggestion:** Return an error from `trigger_checkpoints_sync` rather than panicking, and propagate it so the task exits cleanly. Alternatively, spawn with `task_executor.spawn_critical` so the node terminates if the task panics, making the failure visible.
-

SVC-008: Checkpoint Syncer Background Stream Task Uses Bare

`tokio::spawn` Without Cancellation

- **File:** `crates/botanix-bitcoin-checkpoint/src/syncer.rs:285-288`
 - **Severity:** Medium
 - **Category:** Task lifecycle / Silent task leak
 - **Description:** The ZMQ message consumer is started with a bare `tokio::spawn` with no handle stored and no cancellation token. If this task panics, the failure is swallowed. When the outer `sync` loop exits, the spawned task continues running, holding the ZMQ stream resource. There is no shutdown signal, so a clean shutdown of the sync loop leaves an orphaned task.
 - **Suggestion:** Use a cancellation token and store the `JoinHandle`, or restructure to run both the stream consumer and the sync loop in the same `async` scope using `tokio::select!`.
-

SVC-009: CometBFT RPC Connection Uses Plain HTTP

- **File:** `bin/botanix-reth/src/services/cometbft.rs:11-16`
 - **Severity:** Medium
 - **Category:** Transport security / Unencrypted communication
 - **Description:** The CometBFT RPC factory always uses `http://`. Traffic between the EVM node and the CometBFT consensus layer is transmitted in cleartext. An attacker with network access between these processes can observe or tamper with consensus data.
 - **Suggestion:** Support HTTPS connections by making the scheme configurable. At minimum, warn at startup if the host is not `localhost` / `127.0.0.1`.
-

SVC-010: Federation Config File Read Error Discards Root Cause

- **File:** `bin/botanix-reth/src/services/frost.rs:91-94`
- **Severity:** Medium
- **Category:** Error handling / Configuration

- **Description:** When `load_federation_config_toml()` fails, the original error is discarded with `Err(_) => { ... }` and replaced with a generic "Failed to read federation config file" message. This loses valuable debugging information such as file-not-found, permission denied, TOML parse errors, or invalid public key formats. For a security-critical configuration file that defines federation membership, clear error reporting is essential.
 - **Suggestion:** `let federation_config = load_federation_config_toml(&poa_cfg.federation_config_path).wrap_err("Failed to read federation config file"?);`
-

SVC-011: UTXO Recovery Error Silently Returns `Ok()`

- **File:** `bin/botanix-reth/src/services/recover_utxos.rs:42-43`
 - **Severity:** Low
 - **Category:** Silent failure / Error handling
 - **Description:** When `btc_server_client.recover_missing_utxos(recover_request).await` fails, the error is logged but `Ok()` is returned anyway. A failed UTXO recovery is indistinguishable from a successful one at the call site. The caller in `main.rs:275` uses `?` on the result expecting errors to propagate, but they never will for recovery failures.
 - **Suggestion:** Return `Err` on RPC failure so the caller can decide whether to abort or continue.
-

SVC-012: FROST Protocol Event Channel Has Excessively Large Fixed Buffer

- **File:** `bin/botanix-reth/src/services/network_builder.rs:162`
 - **Severity:** Low
 - **Category:** Resource management
 - **Description:** The FROST protocol event channel is created with a fixed buffer size of 10,000. If the FROST manager consumer falls behind, up to 10,000 protocol events containing cryptographic data (signatures, commitments) will be buffered in memory. Under adversarial conditions where a malicious peer floods FROST protocol messages, this could consume significant memory before backpressure is applied.
 - **Suggestion:** Consider a smaller buffer size (e.g., 256-1024) with proper backpressure handling, or add monitoring for channel utilization.
-

SVC-013: Trusted Peers Added in Two Locations Without Deduplication

- **File:** `bin/botanix-reth/src/services/frost.rs:81-84` and `bin/botanix-reth/src/services/reth.rs:26-30`
- **Severity:** Low
- **Category:** Configuration

- **Description:** Trusted peers from `network_args.trusted_peers` are added to `reth_config.peers.trusted_nodes` in both `load_reth_config()` and `setup_frost()`. If both functions are called with the same config, peers will be duplicated in the trusted nodes list.
 - **Suggestion:** Add trusted peers in only one location. Consolidate all trusted peer management in `setup_frost()`.
-

SVC-014: Database Migration Cleanup Deletes Directory on Any Failure

- **File:** `bin/botanix-reth/src/services/migrator.rs:54-64`
 - **Severity:** Low
 - **Category:** Service lifecycle / Data integrity
 - **Description:** When `migrate_botanix_tables()` fails, the error handler removes the entire botanix database directory with `fs::remove_dir_all()`. If the migration failure is due to a transient issue (disk full, I/O error), recoverable partial migration data is destroyed. If `remove_dir_all` also fails, both errors are propagated but the cleanup error shadows the original.
 - **Suggestion:** Consider a more granular cleanup strategy using a migration-in-progress marker file. Log the original migration error before attempting cleanup.
-

SVC-015: No Validation That `min_signers` Is at Least 2 in Federation Mode

- **File:** `bin/botanix-reth/src/services/frost.rs:56-62`
 - **Severity:** Low
 - **Category:** Configuration
 - **Description:** The FROST configuration validation only checks that `min_signers <= max_signers` but does not validate that `min_signers >= 2`. A `min_signers` value of 0 or 1 would effectively disable the threshold security property of FROST, allowing a single party to produce valid signatures.
 - **Suggestion:** Add a check that `min_signers >= 2` in federation mode. This is a fundamental security invariant for threshold signature schemes.
-

SVC-016: Metrics VersionInfo Contains Hardcoded Placeholder Strings

- **File:** `bin/botanix-reth/src/services/metrics.rs:36-39`
- **Severity:** Low
- **Category:** Metrics / Observability
- **Description:** The `VersionInfo` struct contains hardcoded string literals `"VERGEN_CARGO_FEATURES"`, `"VERGEN_CARGO_TARGET_TRIPLE"`, and `"BUILD_PROFILE_NAME"` instead of actual build-time values. The metrics endpoint reports inaccurate build metadata, which could confuse monitoring and incident response.

- **Suggestion:** Use `env!()` / `option env!()` macros to populate these from Cargo environment variables at compile time.

SVC-017: BTC Fee Rate Conversion May Produce Incorrect Values for Negative/NaN Inputs

- **File:** `crates/botanix-rpc-config/src/botanix_config.rs:300-303`
- **Severity:** Low
- **Category:** Input validation
- **Description:** The fee rate conversion `fee.to_float_in(Denomination::Bitcoin) * 100_000.0` followed by `sat_per_vb.ceil() as u64` has edge cases: a negative fee rate produces 0 (saturating in Rust 2021+); NaN or infinity also produce 0 or MAX. The conversion factor may also be subject to floating-point precision issues.
- **Suggestion:** Add explicit validation that the fee rate is positive and finite before the conversion. Consider using integer arithmetic to avoid floating-point precision issues.

Rust Quality Review

Summary

The custom botanix-reth crates are generally well-structured with good use of `?` and typed errors. However, several critical issues were found: a `RandomSourceProvider` that always returns all-zero bytes, panics in production-path `const fn` code handling financial fee splits, a `println!` call inside a production deserialization path, an infinite-recursion `Clone` implementation, and multiple `unreachable!` macros reachable under database corruption or protocol evolution.

QUAL-001: `RandomSourceProvider` always returns all-zero bytes — not actually random

- **File:** `crates/botanix-authority-rsp/src/lib.rs:25-27`
- **Severity:** Critical
- **Category:** Crypto misuse / Incomplete implementation
- **Description:** `RandomSourceProvider::random_source()` returns `FixedBytes::default()` — all zeros — with a `// TODO: Implement a proper random source.` comment. Tests explicitly assert `assert_eq!(random_bytes, FixedBytes::default())`, verifying the broken behavior rather than correct randomness. A deterministic all-zero random source means leader selection will be predictable, undermining the security model. This is the production

struct, not test code. Combined with EXEC-001 (hardcoded coordinator always index 0), this means neither the randomness source nor the rotation logic is functional.

- **Suggestion:** Implement using a CSPRNG (e.g., `rand::rngs::OsRng`) or a VRF-derived value from the chain state. Remove tests that assert zero-byte behavior. Add a compile-time or runtime guard that prevents this from being used in production until implemented.

QUAL-002: `block_fees_split` uses `.expect()` in a `const fn` called in consensus hot paths

- **File:** `crates/botanix-consensus-common/src/utils.rs:18-34`
- **Severity:** High
- **Category:** Panic
- **Description:** `block_fees_split` is a `const fn` that calculates financial fee distributions for every block. It calls `.checked_mul(50).expect(...)` on a `u128` value. While `u128 * 50` cannot overflow for realistic fee values, the design pattern uses `.expect()` — a panic — as the error handling strategy in a consensus-critical function. `const fn` panics at compile time only when used in const context; when called at runtime with network-supplied values, this produces a runtime panic. Using `checked_mul` to guard overflow then calling `.expect()` defeats the purpose of using the checked variant.
- **Suggestion:** Return a `Result<(u128, u128, u128), &'static str>` or use saturating arithmetic. At minimum, document that the function panics for any non-zero `total_block_fees` near `u128::MAX / 50`.

QUAL-003: `println!` in production deserialization path

- **File:** `crates/botanix-authority-edh/src/extra_data_header.rs:139`
- **Severity:** High
- **Category:** Code quality / Logging violation
- **Description:** Inside `ExtraDataHeader::deserialize()`, which is called on every block header processed by the node, there is a `println!("Error: {:?}", e)` when a malformed aggregate public key is encountered. This violates the project's explicit Clippy rule that disallows `print` macros in production code (causing `make lint-clippy` to fail). Writing to stdout from a production deserialization path is a performance anti-pattern and can flood output under adversarial conditions (malformed blocks).
- **Suggestion:** Replace with `tracing::error!("malformed aggregate public key: {:?}", e)` or simply omit the log since the error is already propagated.

QUAL-004: `unreachable!()` in `remove_staged_header` is reachable under database corruption

- **File:** `crates/botanix-storage/src/provider/database/staged_header.rs:59`

- **Severity:** High
 - **Category:** Panic
 - **Description:** `remove_staged_header` matches on remove count: `0 => Ok(false), 1 => Ok(true), _ => unreachable!()`. The project guidelines explicitly disallow `unreachable!()` in production code. This is not logically unreachable: MDBX allows duplicate-value keys in some table configurations, or a concurrent write transaction could insert a duplicate. If the count is ever 2 or more, the node panics, potentially crashing a validator mid-consensus.
 - **Suggestion:** Replace with `_ => Err(ProviderError::Database(DatabaseError::Other(format!("Unexpected count {} removing staged header", res))))`.
-

QUAL-005: Truncating `as u32` cast on Bitcoin block height

- **File:** `crates/botanix-authority-edh/src/header_ext.rs:175` and `crates/botanix-bitcoin-checkpoint/src/syncer.rs:220`
 - **Severity:** High
 - **Category:** Integer overflow / Truncating cast
 - **Description:** `bitcoin_checkpoint_height as u32` where `bitcoin_checkpoint_height` is `usize` (64-bit on 64-bit systems). Bitcoin's current block height (~900,000) is well within `u32::MAX`, but silent truncating casts are dangerous and not future-proof — the RPC returns `u64` heights. If the function is called with an unexpected value, this silently truncates.
 - **Suggestion:** Use `u32::try_from(bitcoin_checkpoint_height).map_err(|_| BotanixConsensusPackageError::FailedToRetrieveBitcoinCheckpointHeight(...))` to make the cast explicit and error on overflow.
-

QUAL-006: `let _ = writer.write(...)` silently discards partial write in EDH serializer

- **File:** `crates/botanix-authority-edh/src/extra_data_header.rs:106`
 - **Severity:** High
 - **Category:** Error propagation / Silent failure
 - **Description:** In `encode_into_without_signature`, the 20-byte Ethereum address is written via `let _ = writer.write(&block_producer_address_bytes)?`. The `?` correctly propagates `io::Error`, but `write` is not guaranteed to write all bytes — it returns the number of bytes actually written, which is discarded with `let _`. For in-memory `Vec` buffers this is fine, but for any streaming writer the address could be silently truncated, producing a structurally invalid EDH that passes the type system but fails deserialization.
 - **Suggestion:** Replace with `writer.write_all(&block_producer_address_bytes)?`.
-

QUAL-007: `unreachable!()` in `RuntimeVersion::decompress` is reachable on database corruption

- **File:** `crates/botanix-storage/src/models/activation_manager.rs:101`
 - **Severity:** Medium
 - **Category:** Panic
 - **Description:** The `Decompress` implementation for `RuntimeVersion` uses `unreachable! ("passed on wrong value to decompress")` when `value.len() < 4`. This is in the database deserialization path. If the database is corrupted, truncated, or a migration bug stores fewer than 4 bytes, the node panics instead of returning a `DatabaseError`.
 - **Suggestion:** Return `Err(reth_storage_errors::db::DatabaseError::Other("RuntimeVersion value too short".into()))`.
-

QUAL-008: `unreachable!()` in NDD serialization for unknown versions

- **File:** `bin/botanix-reth/src/consensus/comet_bft/non_deterministic_data.rs:200`
 - **Severity:** Medium
 - **Category:** Panic
 - **Description:** The NDD `serialize()` function uses `unreachable!()` for version values other than V0/V1/V2. If a new version is added to the enum but this match arm is not updated, the node will crash at runtime rather than returning a descriptive error.
 - **Suggestion:** Return a `Result` with a descriptive error for unsupported versions.
-

QUAL-009: TOCTOU race in `is_migration_needed` / `path_has_content`

- **File:** `crates/botanix-storage-migrate/src/migrate.rs:42-50`
 - **Severity:** Medium
 - **Category:** Storage correctness / Race condition
 - **Description:** `path_has_content` checks `path.exists()` followed by `path.read_dir()` in two separate filesystem operations. Between these calls, another process could create or delete the directory. Additionally, `entries.count()` consumes the `ReadDir` iterator without checking individual entry errors — if a permission error occurs reading an entry, it is silently skipped, potentially reporting a non-empty directory as empty and skipping a needed migration.
 - **Suggestion:** Use `read_dir()` directly (not preceded by `exists()`), and handle errors from individual directory entries explicitly.
-

QUAL-010: `ommer_reward` has a latent subtraction underflow

- **File:** `crates/botanix-consensus-common/src/calc.rs:122`
- **Severity:** Medium
- **Category:** Integer overflow / Underflow

- **Description:** `((8 + ommer_block_number - block_number) as u128 * base_block_reward) >> 3` — both operands are `u64`. If `block_number > ommer_block_number + 8` (which the function itself does not enforce), `8 + ommer_block_number - block_number` wraps to a very large `u64`, resulting in an astronomically large reward. In the Botanix context (post-merge, no ommers) this function may never be called, but it exists in the library.
- **Suggestion:** Add `debug_assert!(ommer_block_number + 8 >= block_number)` and use `checked_sub` or return `Option<u128>` with a saturating calculation.

QUAL-011: Lock ordering inconsistency may cause deadlock in

`on_finalize_block`

- **File:** `crates/botanix-activation-manager/src/manager.rs:518-564`
- **Severity:** Medium
- **Category:** Potential deadlock
- **Description:** In `on_finalize_block`, `active_version` read lock is acquired at line 518, then `upgrade` write lock at line 525. In `on_prepare_proposal`, `upgrade` read lock is acquired before `active_version` read lock (lines 375-403) — the opposite order. If two threads call these concurrently, a deadlock is possible. Additionally, `active_version` read lock is dropped then immediately re-acquired as a write lock — if another thread holds a read lock between these two operations, it creates a window for livelock. All lock acquisitions throughout the activation manager use `.expect("poisoned lock")` which will panic on lock poisoning.
- **Suggestion:** Establish and document a consistent lock ordering (always acquire `upgrade` before `active_version`, or vice versa). Migrate to `parking_lot::RwLock` which does not poison and has better deadlock detection support.

QUAL-012: `ExtraDataHeader::deserialize` does not validate the version field

- **File:** `crates/botanix-authority-edh/src/extra_data_header.rs:128-154`
- **Severity:** Medium
- **Category:** Serialization / Missing validation
- **Description:** `ExtraDataHeader::deserialize` reads `version` and `chain_version` from the stream but never validates them against the known constants `EXTRA_HEADER_VERSION` (0) and `CHAIN_VERSION` (0). The error variant `ExtraDataHeaderDeserializeError::InvalidVersion` exists in the error enum but is never returned. A block with a future or garbage version will be parsed as if it were valid version 0, potentially allowing a malformed EDH to pass consensus validation.
- **Suggestion:** After reading `version`, add:

```
if version != EXTRA_HEADER_VERSION {
    return Err(ExtraDataHeaderDeserializeError::InvalidVersion);
}
```

Similarly for `chain_version`.

QUAL-013: `BotanixConsensusPackageError::Clone` uses `to_owned()` causing potential infinite recursion

- **File:** `crates/botanix-authority-edh/src/header_ext.rs:81-85`
 - **Severity:** Medium
 - **Category:** Type safety
 - **Description:** The `Clone` implementation calls `self.to_owned()`. `to_owned()` has a blanket impl that calls `clone()`, creating infinite recursion. This would cause a stack overflow if `clone()` is ever called on this error type.
 - **Suggestion:** Either derive `Clone` properly (if inner types support it), or remove the `Clone` implementation and use `Arc<BotanixConsensusPackageError>` where shared ownership is needed.
-

QUAL-014: `expect()` on `MINT_CONTRACT_ADDRESS` parsing sets a bad precedent

- **File:** `crates/botanix-authority-peg/src/mint_validation.rs:22`
 - **Severity:** Medium
 - **Category:** Panics in production
 - **Description:** `MINT_CONTRACT_ADDRESS` is a `Lazy<Address>` that calls `.unwrap()` on parse. While this specific hardcoded address string is valid, using `unwrap()` on a `Lazy` static means any failure panics on first access.
 - **Suggestion:** Use a `const` address value, or add a comment explaining why `unwrap()` is safe here.
-

QUAL-015: `expect()` calls in `snapshot_manager.rs` on `std::sync::RwLock`

- **File:** `bin/botanix-reth/src/consensus/snapshot_manager.rs:369, 395, 495, 591, 600`
 - **Severity:** Medium
 - **Category:** Panics in production / Concurrency
 - **Description:** Multiple `expect("poisoned lock")` calls on `std::sync::RwLock`. A poisoned lock means another thread panicked while holding the lock. The first panic's error message is more informative than "poisoned lock".
 - **Suggestion:** Use `.read().unwrap_or_else(|e| e.into_inner())` to recover from poisoned locks, or switch to `parking_lot::RwLock`.
-

QUAL-016: `add_block_id_if_not_exists` builds a temporary `BTreeSet` but underlying storage remains a `Vec`

- **File:** `crates/botanix-storage/src/models/snapshot.rs:403-411`

- **Severity:** Low
 - **Category:** Logic error / Correctness
 - **Description:** `add_block_id_if_not_exists` creates an ephemeral `BTreeSet` from `block_ids` to check for duplicates, then pushes to `self.block_ids` (a `Vec`). The `BTreeSet` is thrown away. The `Vec` remains unsorted and duplicates can still appear if the separate `add_block_id` method (no duplicate check) is called. Same anti-pattern in `add_chunk_id_if_not_exists`.
 - **Suggestion:** Store as a `BTreeSet` directly, or remove `add_block_id` and document that `add_block_id_if_not_exists` is the only safe insertion method.
-

QUAL-017: `VoteWatcher` uses `.unwrap()` on `RwLock`

- **File:** `crates/botanix-activation-manager/src/vote_tracker.rs:36, 62, 69, 76, 83, 90, 106, 122`
 - **Severity:** Low
 - **Category:** Panics in production
 - **Description:** All `RwLock` operations in `VoteWatcher` use `.unwrap()`. While this is an in-memory vote tracker, panics here would still crash the node.
 - **Suggestion:** Use `parking_lot::RwLock` or handle lock poisoning gracefully.
-

QUAL-018: `main.rs` panics on unsupported command instead of exiting gracefully

- **File:** `bin/botanix-reth/src/main.rs:107`
 - **Severity:** Low
 - **Category:** Panics in production
 - **Description:** `_ => panic!("Only 'node' command is supported in Botanix Reth")` produces an unclear exit instead of a proper error message and exit code.
 - **Suggestion:** Use `eprintln!` + `std::process::exit(1)` or `eyre::bail!`.
-

No Issues Found

The following areas were reviewed across both audits and found to be correct and well-implemented:

- **Unsafe code:** No `unsafe` blocks found in any reviewed crate or binary.
- **NDD deserialization error handling:** Unknown version bytes correctly rejected with `InvalidVersion` error.
- **Bitcoin checkpoint validation in `process_proposal`:** Proposed Bitcoin block hash correctly checked against the local checkpoint chain before accepting the proposal.

- **Aggregate public key consistency:** `agg_pk` from local storage correctly compared against the `agg_pk` embedded in the NDD by the proposer. Mismatches are rejected.
- **Block cache eviction:** LRU cache capped at 5 entries, preventing unbounded memory growth from unapplied proposals.
- **`finalize_block` re-execution path for historical sync:** NDD deserialization failures correctly panic rather than silently proceeding with corrupt state — consistent with the documented panic-on-error strategy.
- **NDD version gating:** NDD V0 correctly restricted to testnet-only historical sync.
- **`commit` block ordering:** Botanix database committed before the reth database, with idempotency guarantee documented.
- **Chain ID validation in `init_chain`:** CometBFT chain ID parsed and asserted to match the EVM chain spec ID before any further initialization.
- **Duplicate outpoint detection in pegin parsing:** `HashSet` deduplication of outpoints within a single mint event is correct.
- **Merkle proof validation:** Merkle proof extraction, txid inclusion check, and root comparison against the block header are correctly implemented.
- **Coinbase maturity check:** Correctly rejects spending of non-mature coinbase outputs.
- **Amount wei/satoshi conversion:** `to_wei` / `from_wei_floor` / `from_wei` conversions are correct; `from_wei_floor` guards against overflow via `MAX_SATOSHI` check.
- **Duplicate pegout ID detection:** `HashSet` check in PSBT validation is correct.
- **Change output count validation:** Limit of at most one change output is correctly enforced.
- **State revert on botanix validation failure:** When a mint-contract check fails, state diff is cleared and only nonce + gas cost are applied. Logic is sound.
- **`block_reward` and `base_block_reward`:** Ethereum block reward calculation is correct and well-tested.
- **Federation config parsing:** TOML parsing is properly error-propagated; DNS fallback for socket addresses is handled correctly. `max_signers` validated against actual number of federation authorities. Node validates its own public key is present.
- **`block_fees_split` arithmetic split logic:** The subtraction-based 10% calculation (`total - 50% - 40%`) is correct and avoids rounding error for the third bucket.
- **Migration table transporter:** Correctly verifies entry counts before and after transport, and commits in the right order. On migration failure, partially written database directory is removed before returning.
- **Bitcoin Checkpoints Chain:** Parent-hash linkage strictly enforced on every `push`. Overflow in window-size arithmetic protected with `checked_add` / `checked_sub`. `ArcSwap`-based read path is correctly lock-free.
- **Checkpoint synchronizer reset-on-fork:** `StaleBlockAdded` path correctly clears chain and resets `last_synced_height`, triggering full re-sync on Bitcoin reorgs.
- **Staged header recovery:** Mechanism for recovering and replaying staged headers after btc-server reconnection is logically correct.

- **DataParser serialization:** Error propagation is correct throughout.
- **BotanixProviderFactory transaction management:** Write transactions are committed or dropped correctly in all `SnapshotWriter` implementations.
- **proto_debug.rs:** No security issues. Truncated debug implementations correctly avoid printing large byte arrays. The subtle `dbg_len - 1` path is guarded correctly by the `max_len == 0` early return.
- **RuntimeVersion ordering:** The `Ord` implementation is correct and the `Compress / Decompress` roundtrip is validated by tests.
- **CLI argument parsing:** `parse_ethereum_address` correctly validates length, hex decoding, and checksum. No panics.
- **VoteWatcher approval-rate logic:** Tracking with `min_validator_count` as floor denominator is sound.
- **Activation Manager design:** Well-structured with proper separation of concerns, comprehensive documentation, and good test coverage. Vote retention/expiration mechanism is sound.
- **BotanixHardfork::from(SpecId):** Defaulting unknown specs to `Prague` is intentional since all Botanix hardforks are active from genesis.
- **Panic hook:** `botanix-utils/src/panic_hook.rs` calls `std::process::exit(1)` after logging, ensuring the process terminates on any panic — appropriate for a validator node.